**PROJECT PROPOSAL**

# Machine Learning Optimization of Social Resource Allocation for Poverty Reduction in Rwanda

Using data to find who needs help most, and sharing limited resources fairly

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# 1. Introduction

A critical challenge in the development stages of Rwanda is poverty level reduction, which requires effective identification of vulnerable community and efficient allocation of limited resources.
It has now become imperative for government institutions, non-government development organizations, and local primary districts to determine where resources are urgently and mostly needed, what appropriate intervention is needed, and the best strategies to allocate available resources.
Machine learning approach provides an efficient approach to analysing vital information such as socio-economic, demographic, employment, education, agricultural, health to identify poverty patterns and effective support system for resource allocation compared to the conventional approach which make use of historical statistics, periodical surveys or manually developed reports based on some defined criteria.

# 2. Why This Matters

### Programs that help poor families

- **VUP:** Rwanda's main program for the poorest families. It gives cash to families who cannot work and pays others for community work.
- **Mutuelle de Santé:** health insurance. The government pays for the poorest families. Coverage grew from about 7% in 2003 to over 88% in 2025.
- **One Cow per Family:** a poor family receives a cow for milk, income and manure.

### How families are chosen

**The old way (Ubudehe).** Since 2000, neighbors and local leaders sorted every family into groups from poorest to richest. The group decided what help the family got. There were problems: people who were not poor tried to get into the poorest group, some poor families were put in the wrong group, and the groups were updated only every three years. In 2023 the government stopped using Ubudehe to choose who receives help.

**The new way (Imibereho).** In February 2024 the government launched the Imibereho Social Registry, a national database about every household. Families can see their record by dialling *195#. By October 2024, 3.2 million households (99%) were registered.

### The new way still makes mistakes

In 2026 the government used the registry to set how much each family pays for Mutuelle de Santé. More than **360,000 households appealed** because they thought they were in the wrong group, and officials said some of the data was wrong. People's lives change (a divorce, a new baby, a lost job), and records can miss these changes.

> **A note.** We could not find any public explanation of how the registry works out a household's group. So we do not try to copy it or fix it. We offer another way to find who needs help, one that could work alongside it.

## 3. Problem Statement

Rwanda has made very considerable and impactful efforts in reducing poverty, and improve socioeconomic conditions. However, differential gaps in education, employment, household income, access to healthcare services, food security, and some other factors still expose some community to poverty.

It is not enough to identify community experiencing poverty, it is also important to determine how to prioritize and optimize the allocation of limited development resources.

Therefore, this project is aimed at developing an intelligent system to predict poverty with optimized resource allocation. This system would identify locations with threatening poverty vulnerability and determine allocation strategy subject to available resources constraints.

## 4. Objectives

### Main objective

Develop and test a machine learning and optimization framework that improves how limited social resources are allocated to reduce poverty in Rwanda.

### Specific Objectives

1.Obtain a socioeconomic dataset with relevant features and pre-process

2.Develop a machine learning model to predict poverty vulnerability or poverty risk score and also predict resource needs for vulnerable community

3.Identify the feature(s) that contribute strongly to poverty prediction

4.Integrate an optimization module for allocation of limited resources subject to needs nad other predefined constraints

5.Develop a user-friendly interactive dashboard for poverty-risk analysis and resource allocation visualization

6.Do a demonstration of the system to show an evidence-based poverty-reduction strategies and to help decision making.

## 5. Research Questions

1. How well can a model guess a household's spending from simple facts like housing, belongings and family?
2. What is the level of accuracy of ML model in predicting poverty vulnerability using the identified factors
3. With the same budget, how much more poverty can we reduce by sharing the money smartly than by sharing it evenly or by just picking the poorest first?

4. What does fairness cost? If we force every district to get help, how much poverty reduction do we give up?
5. Among the socioeconomic factors, which strongly influence poverty vulnerability in Rwanda?
6. How can highly constrained resources be allocated among the locations based on the predicted needs

7. How does the interactive dashboard help in the interpretations of poverty predictions and decision makings in resource allocations?

## 6. The Data and What We Checked

### 6.1 The survey

We use the **EICV7 household survey** (2023-24) from the National Institute of Statistics of Rwanda (NISR). Visitors went to **15,054 households** (62,110 people) in all 30 districts between October 2023 and October 2024. Each household answered questions about its house, belongings, family, health, work and spending. Because the survey also measured spending, we know which households are poor.

### 6.2 What counts as poor

| Group | Yearly spending per adult (January 2024 prices) |
|---|---|
| **Extremely poor** | Below 356,432 Rwf |
| **Moderately poor** | From 356,432 up to 560,127 Rwf |
| **Not poor** | 560,127 Rwf or more |

EICV7 measures poverty in a new way, so its poverty numbers **cannot be compared directly with older surveys**. We use EICV7 only.

### 6.3 We opened the real files and checked

| What we checked | What we found |
|---|---|
| **The 21 data files** | Every file has exactly the rows and columns the documentation says. |
| **Households** | 15,054 households, each with its own ID. Every file links to the main poverty file. |
| **Survey design** | 1,674 village clusters in 30 districts (54 clusters in each district, 72 in each of the 3 Kigali districts). |
| **Poverty lines** | 560,127 and 356,432, as in the NISR report. |
| **Share of people who are poor** | Our count: **27.4% poor and 5.4% extremely poor**. NISR publishes the same numbers. |

| What we checked | What we found |
|---|---|
| **The poverty codes** | 1 = extremely poor, 2 = moderately poor, 3 = not poor. When we worked out the groups ourselves from spending, they matched NISR's group for **every** household. |
| **The columns in our plan** | All 109 columns we named exist in the files. |

### 6.4 Things we fixed after checking

- **Answer columns hide in every file.** The columns quintile, pov_jan, poverty, epov_jan and strata_id are in all 21 files, not just the poverty file. When we join files, we must drop them every time.
- **More columns to avoid.** Some columns are built from spending or payments: ed_foodgov (school feeding), s4bq2 (training cost), s5bq4 and s5bq5a (housing paid in kind), s5cq9a and s5cq12 (water payments), and s5fq8 (VUP work results).
- **Use the right spending column.** sol_jan is adjusted to January 2024 prices. The plain spending column cons1ae would give the wrong poor group for about 1 in 5 households (it matches for only 78.8%).
- **VUP columns work differently than we thought.** In each VUP file, s9d1q1 only tells us about that file's own program. For Financial Services it means the household applied for a loan. A few households appear twice, so we count each household once.
- **Missing answers often mean "none".** Animal counts are empty exactly when the household has no animals, so we fill them with 0. The same goes for details of bad events.
- **District sizes.** Districts have 484 to 486 households each, and the Kigali districts have 644 to 647.

## 7. What We Predict and What We Use

### 7.1 The target: what we predict

There is **one target column**, sol_jan. It is the yearly spending per adult in a household, adjusted to January 2024 prices. We predict this number. If it is below 560,127 the household is poor, and if it is below 356,432 the household is extremely poor.

### 7.2 The clues: what we use to predict

We use **many columns** as clues, not just one. A single clue could not tell us how poor a household is. The short names come from the survey form: s5dq1 means Section 5, part d, question 1. The full list is in our **Column Dictionary** file.

| Clue group | Short names | What they tell us |
|---|---|---|
| **Place** | province, district, ur | Where the household lives, in the city or the countryside |
| **Family** | member, s1q1, s1q2, s1q3y, s1q4 | Family size, sex, age, who is the head, marital status |
| **Education** | s4aq2, s4aq3, s4bq4 to s4bq7 | School level, can read, write and calculate |

| Clue group | Short names | What they tell us |
|---|---|---|
| **Health** | s3q2, s3q7 to s3q12 | Health insurance and disabilities |
| **Phone and internet** | s4bq8, s4bq10, s4bq13a, s4bq15 | Phone, type of phone, computer, internet use |
| **The house** | s5aq1 to s5aq3, s5aq7, s5dq1 to s5dq4 | Type of house, rooms, who owns it, walls, roof, floor |
| **Water, power, toilet** | s5cq1, s5cq3, s5cq14, s5cq16, s5cq21, s5cq22a, s5cq25, s5cq26 | Water source, electricity, lighting, cooking, toilet |
| **Animals** | s7aq4a to s7aq4f | Cattle, goats, sheep, pigs, poultry, other |
| **Bad events** | s5eq1, s5eq2a, s5eq4a | Whether something bad happened, what, and how long to recover |
| **Belongings** | s10bq0, s10bq2 | Which of 27 items the household owns, and how many |
| **Accounts** | s10cq1, s10cq4, s10cq8 | Bank or mobile money account, tontine |
| **Distance** | s5fq3a, s5fq3b, s5fq4 | Time and distance to 15 basic services |

The person file has one row per person, so we first turn it into one row per household (for example, number of children, whether the head can read). The 27 belongings and 15 services each become their own columns. In the end we expect around a hundred or more clues.

### 7.3 Columns we never use

These come from spending, so they give away the answer. Using them would make the model look perfect and be useless in real life.

- quintile, pov_jan, poverty, epov_jan, strata_id (in every file).
- Everything starting with exp, plus cons1, cons1ae, food, Food_at_school and hh_index (a price adjustment).
- The payment columns listed in section 6.4, and the house value and rent columns (s5bq1, s5bq2a, s5bq3a), school fees (s4aq14a to s4aq14h) and item values (s10bq5a to s10bq5c).
- All spending and food files.

### 7.4 Helper columns

hhid joins the files. clust tells us which village cluster a household belongs to. weight and pop_wt say how many households or people in Rwanda a surveyed household stands for. ae is household size counted in adult equivalents (children count for less), and the optimizer uses it.

## 8. How We Will Do It

### Step 1. Prepare the data

- Build one row per household from the files, fill the "none" answers, and drop the columns we never use.
- Check that we get the national poverty rate of 27.4%. If not, we stop and find the problem.

### Step 2. Predict poverty

We try three methods and compare them:

- **A simple belongings score:** count what the household owns and draw a line.
- **A simple straight-line model:** a basic formula using a few clues.
- **LightGBM:** a popular tool that learns patterns from the data by itself. This is our main model.

**How we test.** Households in the same village are similar, so we do not test on neighbors of the households we trained on. We keep whole villages aside for testing. This gives an honest score. We measure:

- how close the guess is to the real spending,
- **missed poor households** (poor families the model does not find),
- **wrongly included households** (non-poor families the model picks).

### Step 3. Explain the guesses

We use a method called SHAP. For each household it shows which clues pushed the guess up or down, such as "mud walls and no electricity make the guess lower." Officials can read this and challenge it.

### Step 4. Share the money

This is the part most teams skip. The computer follows four steps:

1. **Find the need.** For each household, the need is how far its spending is below the poverty line.
2. **Find the benefit.** A transfer closes the need, but never by more than the need itself.
3. **Set the rules.** Stay within the budget. One program per household. Every district gets a fair share.
4. **Let the solver choose.** A solver program (PuLP) picks the households that close the most need.

**A small example.** The money covers 3 households, and each receives 100,000 Rwf:

| Household | District | Need | Benefit |
|---|---|---|---|
| **B** | X | 300,000 | 100,000 |
| **D** | X | 250,000 | 100,000 |
| **C** | X | 90,000 | 90,000 |
| **A** | Y | 40,000 | 40,000 |
| **E** | Y | 10,000 | 10,000 |

- **No rules:** the best choice is B, D and C. This closes 290,000, but district Y gets nothing.
- **Rule "every district gets at least one":** the best choice is B, D and A. This closes 240,000.

The fairness rule costs 50,000. We will show this trade-off to the judges. In the real project the solver does the same for thousands of households and chooses between programs. For VUP we use the average yearly amounts from the NISR VUP report: Direct Support about 128,572 Rwf, Classic Public Works about 87,994, Expanded Public Works about 149,422, and Financial Services loans

about 100,000. We assume a transfer raises spending franc for franc. That is the best case, so we also test smaller effects.

> **In one line.** Choose who gets help so that the most need is closed, without going over the budget, and while following the fairness rules.

### Step 5. Compare ways of giving help

We test each way on households the model has not seen, and score them with their real spending:

| Way of giving help | How households are chosen |
|---|---|
| **Even spread** | Same share to every district, households picked at random. |
| **Poorest first** | The highest predicted need first, until the money runs out. |
| **Smart sharing** | The solver closes the most need within the rules. |
| **Smart sharing and fair** | The same, with stricter rules for every district and for the extremely poor. |
| **Perfect knowledge** | The solver, but using real spending. This is the best that could ever be done. |

### What today's help looks like in the survey

In the survey, about **331,000 households** (roughly 1 in 10) receive Direct Support, public works or nutrition support. These households are not mostly poor:

| Measure | Result |
|---|---|
| **Poor households that receive help** | 14.7% |
| **Extremely poor households that receive help** | 17.2% |
| **Non-poor households that receive help** | 8.6% |
| **Households receiving help that are not poor** | 65.7% |

The VUP report finds that 59% of its beneficiaries are not poor, so both sources point the same way. Be careful with these numbers. The programs are small: about 1 in 10 households gets help but about 1 in 4 is poor, so most poor households get nothing simply because the money runs out. Some beneficiaries may also have left poverty because of the help. So we compare today's help with our methods **at the same number of households**, not by raw error rates.

### Step 6. Check fairness

We check results by province, city or countryside, district, sex of the household head and disability. We draw a chart of how much poverty reduction we give up for more fairness.

### Step 7. Build a simple dashboard

An official sets a budget and a program, then sees who is chosen, the reason for each choice and the fairness report.

## 9. Tools

Everything is in Python. The five core tools are enough:

| Tool | What it does |
|---|---|
| **pandas** | Cleans and joins the data. |
| **LightGBM** | The main prediction model. |
| **SHAP** | Explains each guess. |
| **PuLP** | Chooses who gets help within the rules. |
| **Streamlit** | The dashboard. |

We use GitHub to share code and scikit-learn for the simple models and testing. The CSV files and the Column Dictionary help the team open and understand the data.

## 10. Plan

| Phase | What we do | Result |
|---|---|---|
| **0. Setup** | Confirm data access, judging rules and deadline. Share out the work. | Access and shared folder |
| **1. Data** | Build the household table, drop the banned columns, check the 27.4% rate. | Clean table |
| **2. Predict** | Simple methods and LightGBM, village-based testing, SHAP. | Tested model |
| **3. Share money** | Set costs, build the solver, add fairness rules, compare ways. | Comparison |
| **4. Show** | Dashboard, report and slides. | Final package |
| **5. Review** | Check every number and practice the presentation. | Ready to present |

## 11. Deliverables

- A clean household table and the code to make it again.
- A tested model, with its mistakes counted and compared to simple methods.
- A program that shares a budget with fairness rules.
- A comparison of the ways of giving help, and what fairness costs.
- Simple explanations and a fairness report.
- A dashboard, a written report and a presentation.

## 12. Risks

| Risk | What we do about it |
|---|---|
| **We cannot get the data in time** | Ask for access now. As a backup, use the published tables in the NISR reports. |
| **The model sees the answer by accident** | Drop the banned columns from every file. If the score looks too good, we look for a leak. |
| **The score looks better than reality** | Test on whole villages kept aside. |
| **Few households per district (about 486)** | Show how uncertain the results are. Do not make claims about single districts. |
| **Money does not turn into spending one for one** | Say it clearly and test smaller effects. |
| **Not enough time** | Build a small version first: one program, one model, a simple dashboard. Then add more. |

## 13. What This Project Cannot Do

- **Knowing who is poor is not the same as knowing who gains most from help.** Without data from a real program test, we assume poorer families gain more.
- **The VUP survey only covers people who already get help,** so it cannot show who is missed. We train on the main survey instead.
- **One moment in time.** The survey covers October 2023 to October 2024, so we cannot test how a household changes over time.
- **We cannot see the registry's own data or method,** so we cannot say our clues match its clues.
- **Small districts.** About 486 households per district limits what we can say about a district.
- **The poverty measure is an estimate** and has its own errors.
- **Facts that differ by source.** Reports disagree on the new Mutuelle prices. We will confirm them with RSSB before using them as costs.

## 14. Using the Data Responsibly

- Use only the anonymous public files. Never try to find out who a household is.
- Never share the raw data. Share code and summary results only.
- Credit NISR as the source (EICV7 2023-2024).
- Our tool helps people decide. It does not decide for them. Every household it flags must be possible to check and appeal.
- Report openly where the model does worse.

## 15. Sources

### Social protection and targeting

- Social protection strategies and the new Ubudehe categories (FES Rwanda)
- What next after govt scraps Ubudehe categories? (The New Times)
- Govt clarifies on reforming identification of needy people (Africa-Press Rwanda)
- Imibereho Dynamic Social Registry launched (MINALOC)
- Over 80% of people with disabilities identified for needed support (AllAfrica)
- Social registries and digital platforms for inclusive service delivery (World Bank)

### Mutuelle de Santé

- Mutuelle reforms ease payment burden with instalment option (The New Times)
- Misclassified under Mutuelle de Santé? Here's what to do (AllAfrica)
- Rwanda resolves 340,000 appeals over new Mutuelle contribution categories
- Rwanda hikes health insurance contributions but poorest will pay zero (KT Press)

### Data and survey documents

- NISR microdata catalog: EICV7 2023-24, Cross-Sectional Sample (catalog 119)
- NISR: EICV7 survey revealed significant decline in poverty

NISR (2025). EICV7 VUP Thematic Report. NISR also provided the EICV7 data files, questionnaire and documentation that we checked our plan against.

## Appendix: Simple Glossary

| Word | Simple meaning |
|---|---|
| **Target** | The one column we want to predict (sol_jan). |
| **Feature (clue)** | A column we use to make the prediction. |
| **Ubudehe** | The old way of sorting families by wealth, through village discussions. |
| **Imibereho registry** | The national household database that replaced Ubudehe for choosing who gets help. |
| **VUP** | Rwanda's main program for the poorest families (cash and community work). |
| **Mutuelle de Santé** | Rwanda's community health insurance. |
| **NISR** | National Institute of Statistics of Rwanda. It runs the household surveys. |
| **EICV7** | The seventh living conditions survey (2023-24). Our data source. |
| **Adult equivalent** | A way of counting household size where a child counts for less than an adult. |
| **Poverty line** | The spending level below which someone is called poor: 560,127 Rwf a year per adult. |
| **Inclusion error** | Helping a family that is not poor. |
| **Exclusion error** | Missing a family that is poor. |

| Word | Simple meaning |
|---|---|
| **Data leak** | When the model is given something that gives away the answer, so it looks better than it really is. |
| **Survey weight** | How many households in the country one surveyed household stands for. |
| **SHAP** | A method that shows which clues pushed a guess up or down. |
| **Solver** | A program that finds the best choice under a budget and rules. |